

2. The plane  $\Pi$  passes through the point  $A$  and is perpendicular to the vector  $\mathbf{n}$

Given that

$$\overrightarrow{OA} = \begin{pmatrix} 5 \\ -3 \\ -4 \end{pmatrix} \quad \text{and} \quad \mathbf{n} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

where  $O$  is the origin,

- (a) find a Cartesian equation of  $\Pi$ .

(2)

With respect to the fixed origin  $O$ , the line  $l$  is given by the equation

$$\mathbf{r} = \begin{pmatrix} 7 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -5 \\ 3 \end{pmatrix}$$

The line  $l$  intersects the plane  $\Pi$  at the point  $X$ .

- (b) Show that the acute angle between the plane  $\Pi$  and the line  $l$  is  $21.2^\circ$  correct to one decimal place.

(4)

- (c) Find the coordinates of the point  $X$ .

(4)

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4. The line  $l$  has equation

$$\frac{x+2}{1} = \frac{y-5}{-1} = \frac{z-4}{-3}$$

The plane  $\Pi$  has equation

$$\mathbf{r} \cdot (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = -7$$

Determine whether the line  $l$  intersects  $II$  at a single point, or lies in  $II$ , or is parallel to  $II$  without intersecting it.

(5)



4.

**All units in this question are in metres.**

A lawn is modelled as a plane that contains the points  $L(-2, -3, -1)$ ,  $M(6, -2, 0)$  and  $N(2, 0, 0)$ , relative to a fixed origin  $O$ .

- (a) Determine a vector equation of the plane that models the lawn, giving your answer in the form  $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$  (3)

- (b) (i) Show that, according to the model, the lawn is perpendicular to the vector  $\begin{pmatrix} 1 \\ 2 \\ -10 \end{pmatrix}$
- (ii) Hence determine a Cartesian equation of the plane that models the lawn.
- (4)**

There are two posts set in the lawn.

There is a washing line between the two posts.

The washing line is modelled as a straight line through points at the top of each post with coordinates  $P(-10, 8, 2)$  and  $Q(6, 4, 3)$ .

- (c) Determine a vector equation of the line that models the washing line. **(2)**
- (d) State a limitation of one of the models. **(1)**

The point  $R(2, 5, 2.75)$  lies on the washing line.

- (e) Determine, according to the model, the shortest distance from the point  $R$  to the lawn, giving your answer to the nearest cm. (2)

Given that the shortest distance from the point  $R$  to the lawn is actually 1.5 m,

- (f) use your answer to part (e) to evaluate the model, explaining your reasoning. (1)



9. An octopus is able to catch any fish that swim within a distance of 2 m from the octopus's position.

A fish  $F$  swims from a point  $A$  to a point  $B$ .

The octopus is modelled as a fixed particle at the origin  $O$ .

Fish  $F$  is modelled as a particle moving in a straight line from  $A$  to  $B$ .

Relative to  $O$ , the coordinates of  $A$  are  $(-3, 1, -7)$  and the coordinates of  $B$  are  $(9, 4, 11)$ , where the unit of distance is metres.

- (a) Use the model to determine whether or not the octopus is able to catch fish  $F$ . (7)
- (b) Criticise the model in relation to fish  $F$ . (1)
- (c) Criticise the model in relation to the octopus. (1)

4. Part of the mains water system for a housing estate consists of water pipes buried beneath the ground surface. The water pipes are modelled as straight line segments. One water pipe,  $W$ , is buried beneath a particular road. With respect to a fixed origin  $O$ , the road surface is modelled as a plane with equation  $3x - 5y - 18z = 7$ , and  $W$  passes through the points  $A(-1, -1, -3)$  and  $B(1, 2, -3)$ . The units are in metres.

(a) Use the model to calculate the acute angle between  $W$  and the road surface.

(5)

A point  $C(-1, -2, 0)$  lies on the road. A section of water pipe needs to be connected to  $W$  from  $C$ .

(b) Using the model, find, to the nearest cm, the shortest length of pipe needed to connect  $C$  to  $W$ .

(6)

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8. A gas company maintains a straight pipeline that passes under a mountain.

The pipeline is modelled as a straight line and one side of the mountain is modelled as a plane.

There are accessways from a control centre to two access points on the pipeline.

Modelling the control centre as the origin  $O$ , the two access points on the pipeline have coordinates  $P(-300, 400, -150)$  and  $Q(300, 300, -50)$ , where the units are metres.

- (a) Find a vector equation for the line  $PQ$ , giving your answer in the form  $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ , where  $\lambda$  is a scalar parameter.

(2)

The equation of the plane modelling the side of the mountain is  $2x + 3y - 5z = 300$

The company wants to create a new accessway from this side of the mountain to the pipeline.

The accessway will consist of a tunnel of shortest possible length between the pipeline and the point  $M(100, k, 100)$  on this side of the mountain, where  $k$  is a constant.

- (b) Using the model, find

- (i) the coordinates of the point at which this tunnel will meet the pipeline,
- (ii) the length of this tunnel.

(7)

It is only practical to construct the new accessway if it will be significantly shorter than both of the existing accessways,  $OP$  and  $OQ$ .

- (c) Determine whether the company should build the new accessway.

(2)

- (d) Suggest one limitation of the model.

(1)



6. The line  $l_1$  has equation  $\mathbf{r} = \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$  where  $\lambda$  is a scalar parameter.

The line  $l_2$  is parallel to  $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$

(a) Show that  $l_1$  and  $l_2$  are perpendicular.

(2)

The plane  $\Pi$  contains the line  $l_1$  and is perpendicular to  $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$

(b) Determine a Cartesian equation of  $\Pi$

(2)

(c) Verify that the point  $A(3, 1, 1)$  lies on  $\Pi$

(1)

Given that

- the point of intersection of  $l_1$  and  $l_2$  has coordinates  $(2, 3, 2)$
- the point  $B(p, q, r)$  lies on  $l_2$
- the distance  $AB$  is  $2\sqrt{5}$
- $p, q$  and  $r$  are positive integers

(d) determine the coordinates of  $B$ .

(6)



6. The surface of a horizontal tennis court is modelled as part of a horizontal plane, with the origin on the ground at the centre of the court, and

- $\mathbf{i}$  and  $\mathbf{j}$  are unit vectors directed across the width and length of the court respectively
- $\mathbf{k}$  is a unit vector directed vertically upwards
- units are metres

After being hit, a tennis ball, modelled as a particle, moves along the path with equation

$$\mathbf{r} = (-4.1 + 9\lambda - 2.3\lambda^2)\mathbf{i} + (-10.25 + 15\lambda)\mathbf{j} + (0.84 + 0.8\lambda - \lambda^2)\mathbf{k}$$

where  $\lambda$  is a scalar parameter with  $\lambda \geq 0$

Assuming that the tennis ball continues on this path until it hits the ground,

- (a) find the value of  $\lambda$  at the point where the ball hits the ground. (2)

The direction in which the tennis ball is moving at a general point on its path is given by

$$(9 - 4.6\lambda)\mathbf{i} + 15\mathbf{j} + (0.8 - 2\lambda)\mathbf{k}$$

- (b) Write down the direction in which the tennis ball is moving as it hits the ground. (1)
- (c) Hence find the acute angle at which the tennis ball hits the ground, giving your answer in degrees to one decimal place. (4)

The net of the tennis court lies in the plane  $\mathbf{r} \cdot \mathbf{j} = 0$

- (d) Find the position of the tennis ball at the point where it is in the same plane as the net. (3)

The maximum height above the court of the top of the net is 0.9 m.

Modelling the top of the net as a horizontal straight line,

- (e) state whether the tennis ball will pass over the net according to the model, giving a reason for your answer. (1)

With reference to the model,

- (f) decide whether the tennis ball will actually pass over the net, giving a reason for your answer. (2)

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- 6.** The drainage system for a sports field consists of underground pipes.

This situation is modelled with respect to a fixed origin  $O$ .

According to the model,

- the surface of the sports field is a plane with equation  $z = 0$
- the pipes are straight lines
- one of the pipes,  $P_1$ , passes through the points  $A(3, 4, -2)$  and  $B(-2, -8, -3)$
- a different pipe,  $P_2$ , has equation  $\frac{x-1}{2} = \frac{y-3}{4} = \frac{z+1}{-2}$
- the units are metres

- (a) Determine a vector equation of the line representing the pipe  $P_1$  (2)
- (b) Determine the coordinates of the point at which the pipe  $P_1$  meets the surface of the playing field, according to the model. (2)

Determine, according to the model,

- (c) the acute angle between pipes  $P_1$  and  $P_2$ , giving your answer in degrees to 3 significant figures, (3)
- (d) the shortest distance between pipes  $P_1$  and  $P_2$  (5)

[illegible]